Biodiesel as an Alternative Fuel in Siemens Dry Low Emissions Combustors: Atmospheric and High Pressure Rig Testing

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Atmospheric and high pressure rig tests were conducted to investigate the feasibility of using biodiesel as an alternative fuel to power industrial gas turbines in one of the world's leading dry low emissions (DLE) combustion systems, the SGT-100. At the same conditions, tests were also carried out for mineral diesel to provide reference information to evaluate biodiesel as an alternative fuel. In atmospheric pressure rig tests, the likelihood of the machine lighting was identified based on the measured probability of the ignition of a single combustor. Lean ignition and extinction limits at various air temperatures were also investigated with different air assist pressures. The ignition test results reveal that reliable ignition can be achieved with biodiesel across a range of air mass flow rates and air fuel ratios (AFRs). In high pressure rig tests, emissions and combustion dynamics were measured for various combustor air inlet pressures, temperatures, combustor wall pressure drops, and flame temperatures. These high pressure rig results show that biodiesel produced less NO_x than mineral diesel. The test results indicate that the Siemens DLE combustion system can be adapted to use biodiesel as an alternative fuel without major modification. [DOI: 10.1115/1.3204617]

1 Introduction

Renewable fuels are of increasing interest for many applications including gas turbines [1,2] due to environmental issues, fossil fuel conservation, and energy price increases. Biofuels including biogas, bioethanol and biodiesel are renewable and environment friendly in many ways such as being carbon neutral and in theory producing zero carbon dioxide (CO2), but the production of biofuels should avoid competition with food production. The current work focuses on the technical issue of biofuels in terms of operability and emissions.

Biodiesel is produced by the process of transesterification of vegetable oil, animal oil/fats, tallow and waste cooking oil to form fatty acid methyl esters (FAME). The advantage of treating the vegetable oil in this way is that the resulting oil has properties that are significantly closer to those of mineral diesel than in the case for the original feedstock. The precise properties of the fuel produced in this way are dependent on the original feedstock [3,4]. Oils derived straight from crops such as rapeseed, palm and soybean have the greatest potential; however, there are issues with land availability and cost. Waste vegetable oils from restaurants and animal oil/fat from the food industry represent the economic way to produce biodiesel.

Biodiesel fuels have been widely used in internal combustion engines, either pure biodiesel BD100 or blended with diesel as BD5 (biodiesel 5%, diesel 95%), BD10 (biodiesel 10%, diesel 90%). Lapuerta et al. [5] used pure biodiesel fuels in a diesel engine and found that pure biodiesel, compared with the reference mineral diesel, resulted in a slight increase in fuel consumption, very slight differences in NO_x emissions, and sharp reductions in total hydrocarbon emissions, smoke opacity and particle emissions. Biodiesel is indeed an environmentally friendly fuel when applied to internal combustion engines.

To the authors' best knowledge, biodiesel was not widely used in industrial gas turbines, but some test experience was built up on heavy duty gas turbines [1,2], small gas turbines [6], microturbines [7,8] and simulated gas turbine burner tests [9,10]. Comparable NO_x emissions were found in the work on heavy duty gas turbines [1,2] and on small gas turbines [6]. Chiang et al. [8] reported carbon deposits on the fuel nozzle burning biodiesel, and Bolszo et al. [7] found that engine cold start becomes a significant challenge under more extreme conditions. The major concerns using biodiesel to power industrial gas turbine are operability, NO_x emission, materials, component life, fuel storage [11], cold start [7], and carbon deposits [8].

Siemens Industrial Turbomachinery Ltd. was involved in the recently finished European project, Alternative Fuels for Industrial Gas Turbines (AFTUR), to investigate the feasibility of using biodiesel as an alternative fuel to power industrial gas turbines. Siemens conducted both atmospheric and high pressure rig tests in the SGT-100 using one of the world's leading DLE combustion systems. Reference data were obtained by carrying out the same tests with mineral diesel.

In an atmospheric pressure rig, extensive ignition tests were conducted, to investigate engine start reliability, lean ignition, and extinction limits to establish the biodiesel's operability in gas turbines. In high pressure rig testing, emissions and combustion dynamics were investigated. The biodiesel selected for this study was supplied by C Zero Energy Ltd. (Coventry, UK), produced from recycled cooking oil, and was consequently derived from a mixture of feedstocks. All tests were with the standard DLE SGT-100 burner, combustor, and transition duct, and 100% biodiesel (BD100) was used. The reference mineral diesel was BS 2869 Class A2/D.

2 Siemens DLE Combustion System

Siemens Industrial Turbomachinery Ltd. is the manufacturer of small industrial gas turbines from the SGT-100 at approximately 5 MW to the SGT-400 at 13.4 MW, which have now accumulated more than 3 million hours of experience. The SGT-300 and SGT-400 are offered commercially for less than 10 ppmv NO_x

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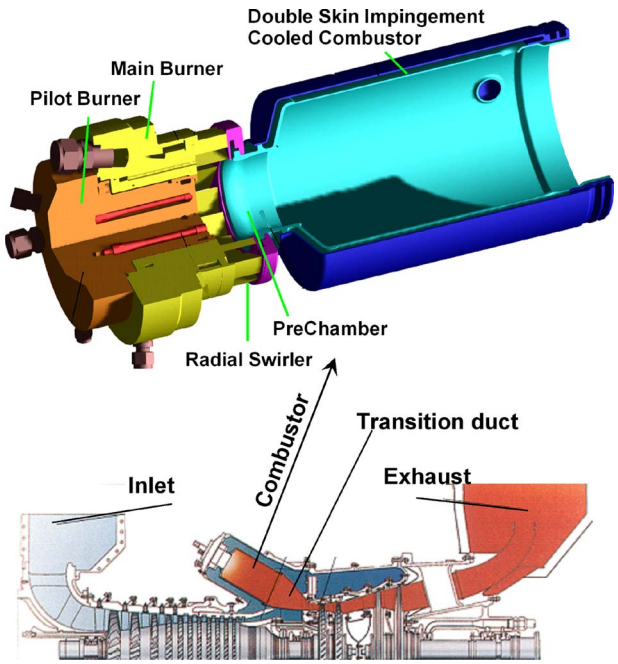

Fig. 1 Siemens DLE combustor construction

(at 15%O_2) emissions for certain markets, for example, the SGT-300, operating at the University of New Hampshire, USA.

The dual fuel DLE combustion system is designed as a reverse-flow tuboannular type, as shown in Fig. 1. The combustor consists of three main sections: (1) the pilot burner, which houses the pilot fuel galleries and injectors for both gaseous and liquid fuels; (2) the main burner, which houses the main air swirler and main gas and liquid fuel injection systems; and (3) the combustor, which includes the prechamber, and is of a double skin construction, cooled through impingement cooling. At the downstream of the combustor, a transition duct is used to condition the flow from the circular combustor exit to a sector of the turbine entry annulus.

In order to cover all the power range, rather than having a single combustor design and applying different numbers of combustors to each engine type, the combustors have been scaled. Figure 2 shows the DLE combustor family. The SGT-100, 300, and 400 have 6 combustors per engine, and the SGT-200 has 8.

A schematic of the combustion concept is shown in Fig. 3. The main combustion air enters through a single radial swirler at the head of the combustor. The flow then turns through a right angle into the prechamber, followed by a sudden expansion into the combustion chamber.

The swirl number is sufficiently high to induce a vortex breakdown reverse flow zone along the axis. This is termed as the internal reverse flow zone. In the concept, this reverse flow zone

Fig. 2 Siemens DLE combustor family

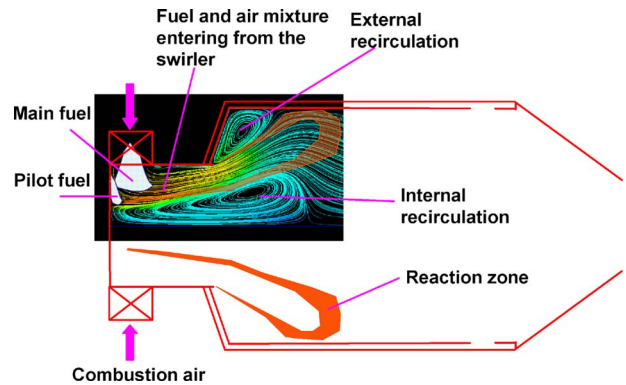

Fig. 3 Siemens DLE combustion concept

remains attached to the surface of the pilot burner, thereby establishing a firm aerodynamic base for flame stabilization. In the wake of the sudden expansion, an external reverse flow zone is established. The flame is stabilized in the shear layers around the internal and external reverse flow zones.

The fuel, either gas or liquid, is introduced in two stages, as shown in Fig. 4: the main, which results in a high degree of premixedness, and hence, low NO_x emissions, and the pilot, which steadily increases as the load demand decreases in order to ensure flame stability. The pilot is arranged, such that, as the pilot fuel split increases, the fuel is biased towards the axis of the combustor.

3 Capability of the DLE System for Fuel Flexibility

Siemens has a continuous policy of expanding its fuel capability. As well as the above mentioned DLE combustion system, Siemens also has conventional combustion systems, which can burn various fuels in the Wobbe index range from 4 MJ/m^3 to 70 MJ/m^3. In this section, the fuel flexibility for the DLE system is described.

The Siemens standard DLE combustion system is capable of burning conventional fuels with a Wobbe index from 37 MJ/m^3 to 49 MJ/m^3 with guaranteed NO_x, CO, and UHC emissions. With modification when necessary, the DLE system is also capable of burning fuels with a lower Wobbe index. For example, the standard SGT-300 operates in Libya with a gas having a Wobbe index of about 32 MJ/m^3, modified SGT-400 engines operate at Yadana in the Andaman Sea with a Wobbe index of 28 MJ/m^3, and at Kandhkot, Pakistan with 32 MJ/m^3.

Siemens is interested in expanding their fuel flexibility with alternative fuels such as syngas, biogas, bioethanol, and biodiesel. High pressure rig tests have been completed with biogases, simulated by diluting natural gas with inert gases and syngases, mixtures of natural gas with hydrogen, carbon monoxide, and inerts.

The current liquid fuel capability for the Siemens DLE system includes mineral diesels and kerosenes. As part of its policy to increase liquid fuel flexibility, Siemens was involved in the Euro-

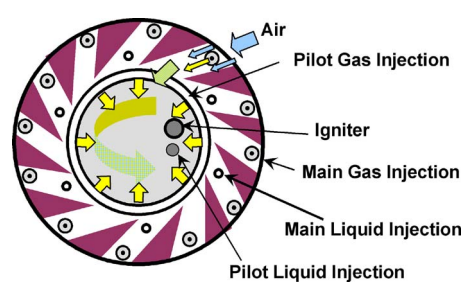

Fig. 4 Conceptual details of fuel injectors and air paths

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pean project AFTUR to investigate the feasibility of using biodiesel as an alternative fuel to power industrial gas turbines. Siemens conducted both atmospheric and high pressure rig tests in the SGT-100 DLE combustion system to investigate ignition, emission, and combustion dynamics using biodiesel as an alternative fuel. Reference data were obtained by carrying out the same tests with mineral diesel. Valuable experience has been gained on biodiesel, and the test results are the basis for this paper.

4 Rig Tests Using Biodiesel to Fuel a Standard SGT-100 Combustor

4.1 Ignition Tests in an Atmospheric Rig. In the atmospheric rig testing, extensive ignition testing of biodiesel has been conducted to investigate the feasibility of using it as a fuel to power industrial gas turbines. For all the tests performed in the atmospheric rig, 100% of the fuel was supplied from the pilot fuel nozzle. Two series of tests were performed to identify the key properties of the proposed fuel. In the first set of tests, an attempt was made to identify the likelihood of the engine lighting, based on the measured probability of the ignition of a single combustor. The second series of tests investigated the lean ignition and extinction limits at various air temperatures and at different air assist pressures.

In order to assess the engine start reliability operating with biodiesel, it is necessary to determine the ignition reliability of a single combustor, and then convert this into an overall probability for the engine lighting. The probability of a single combustor lighting is easily determined by counting the number of sparks required to initiate combustion during the atmospheric rig testing. The probability of ignition is then just the reciprocal of the number of sparks that was taken to ignite the combustor. In practice, the experiment was repeated three times, and an average number of sparks was taken to ensure the consistency, and compensate for noise factors, for example, in the spark.

An algorithm was developed to convert the single combustor ignition probability into a probability for lighting the engine. The method consists of calculating the probability of various permutations of combustors lighting on the first round, based on a given fixed probability of an individual combustor lighting P_i. Any probability of any combination from this first round $P(\text{of } k \text{ from } n)=P_i^k(1-P_i)^{n-k}n!/[(n-k)!k!]$ is then multiplied by the probabilities for lighting various combinations of the remaining unlit combustors. The same process is repeated in the third round. All probabilities for sequences, which lead to all combustors lit, are added together, apart from those, which are excluded by the time-out rules built into the control algorithm. This gives the final probability of a successful system light for a given fixed individual combustor light probability. A number of calculations are then run with different fixed probabilities for the individual combustor, and a polynomial fit is applied to model the system response over the interesting high light reliability end of the curve.

The lean ignition limit was found by activating the igniter, and then gradually increasing the fuel flow rate until ignition took place. Ignition was deemed to have occurred when the flame persisted without the presence of a spark. The effect of air assist pressure and air temperature on lean ignition limit was investigated. The lean extinction limit was determined by taking the stable flame at each of the air flow rates studied, and reducing the amount of fuel until extinction occurred. At this point, the air and fuel flow rates were recorded. Each test was repeated three times to ensure the consistency, and compensate for the noise factors such as fluctuations in the air or fuel flow rates.

4.2 Emissions and Dynamics Tests in a High Pressure Rig. High pressure rig tests were performed in one of the high pressure facilities at Firth Road, Lincoln, of Siemens Industrial Turbomachinery Ltd. The maximum pressure, temperature, and air mass flow rate are 20 bar, 873 K, and 5 kg/s, respectively.

The high pressure rig tests investigated the behavior of the fuel injection system, which directly affects the degree of premixedness of the air and fuel mixture, as the operating condition was altered under a biodiesel fuel operation. This was achieved by making measurements of the bulk emissions at the exit of the combustor, and then inferring the impact on the spray from the observed change.

The key parameters influencing the quality of the atomization are believed to be inlet air pressure, temperature, mass flow rate, and flame temperature. A different air mass flow rate results in a different combustor wall pressure drop, which is a key parameter to control air fuel mixing and flame stability. As a consequence, the variation in these parameters formed the basis of the present testing for biodiesel. At the same operating conditions, reference tests were also performed for a mineral diesel operation. During the tests, combustion dynamics were also recorded. For all the tests, 100% of the fuel was supplied from main fuel nozzles with no fuel supplied from the pilot fuel nozzle. The pilot fuel nozzle is an air assist injector, and the main nozzle is a pressure swirl injector.

Table 1 The comparison of fuel properties of biodiesel with those of diesel

Property	Diesel	Biodiesel
Low heating value (MJ/kg)	42.397	37.414
Density at 15°C (kg/m^3)	867.8	832.2
Carbon (% mass)	87.1	76.8
Hydrogen (% mass)	12.7	12.1
Oxygen (% mass)	0.0	10.6
Nitrogen (mg/kg)	335.0	25.0
Kinematic viscosity at 40°C (mm^2/s)	3.45	4.91
Kinematic viscosity at 50°C (mm^2/s)	–	2.67

4.3 Biodiesel Supply and Its Properties. The biodiesel selected for this study was composed of recycled cooking oil. The fuel samples of biodiesel and diesel were collected before and after the test, and were sent to a laboratory for analysis. The comparison of the important properties of the two fuels is listed in Table 1. One can see that biodiesel has a lower heating value, contains oxygen, and in particular, the kinematic viscosity is about 42.3% higher than that of diesel at a temperature of 40°C. When the fuel temperature is increased to 50°C, the biodiesel's kinematic viscosity is reduced to 54.4% of its value at 40°C.

Tate et al. [3,4] studied the fuel properties of three biodiesel fuels from different sources, methyl esters (ME) from canola and soy, and ethyl esters (EE) from fish oil. All three of these biodiesel fuels have much higher kinematic viscosity than diesel, and the viscosity of biodiesel exponentially decays as temperature increases, with significant change with the temperature at the range of 20–100°C. The boiling points of the three biodiesel fuels are also higher than diesel.

Wilson et al. [11] showed that their four kinds of biodiesel fuels have higher viscosity than diesel, and the carbon content varies from 76.8% to 77.52%. The hydrogen content varies from 12.59% to 12.77%, oxygen content varies from 9.37% to 9.94%, and the nitrogen content varies from 0.34% to 0.38%. One can see that the biodiesel selected for the current work has less nitrogen content than that of Wilson et al. [11].

Panchasara et al. [9] also reported that the viscosity of biodesel is 45–55% higher, compared with that of mineral diesel. The higher fuel viscosity of biodiesel could result in poor atomization, which will be quantified later on by droplet size predictions. Surface tension is required in order to evaluate the droplet size of the fuels. It was calculated for both fuels at different fuel pressures by

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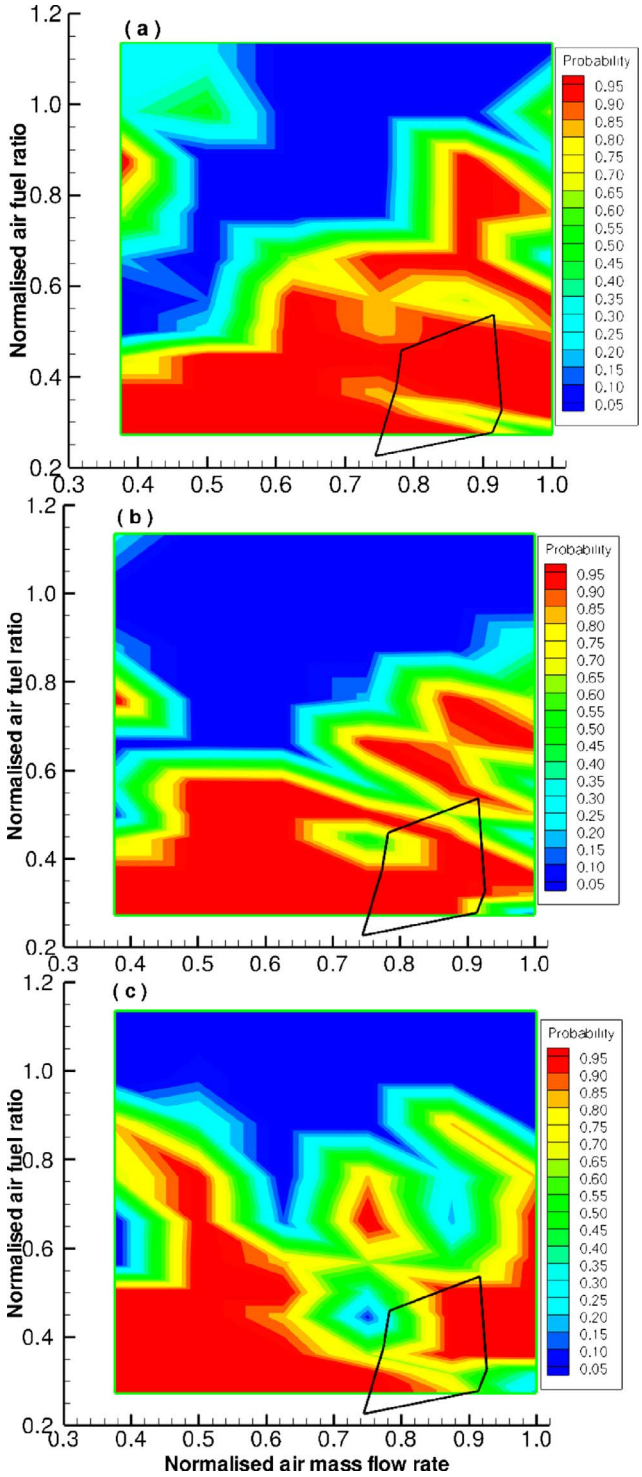

Fig. 5 Probability of engine lighting, fueled with biodiesel and calculated from single combustor tests for three different air assist pressures: (*a*) 2.31 bar, (*b*) 2.72 bar, and (*c*) 3.21 bar

the method of Odgers and Krestschmer [12]. The calculation showed that both fuels have comparable surface tension, which is also confirmed by Panchasara et al. [9].

5 Test Results and Discussion

5.1 Ignition. Figure 5 shows the predicted probabilities of the engine lighting with biodiesel, based on the measured probability for the ignition of a single combustor. Also, three levels of air assist pressure are shown in the figure. The black quadrilateral overlaying these plots shows the predicted extreme limits of the SGT-100 light up window, which incorporates the combustor to combustor variation as the engine lights for biodiesel. The prediction for the light up window of diesel is very close to that of biodiesel.

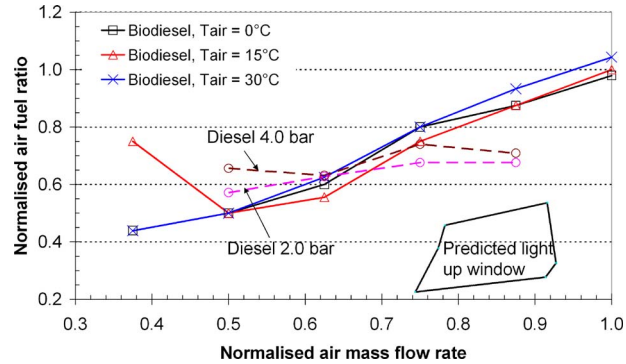

Fig. 6 Lean ignition limit for SGT-100 burning biodiesel at different ambient air temperatures at an air assist pressure of 2.72 bar, and the comparison with diesel

At low air mass flow and low air fuel ratio, it seems that air assist pressure has little impact on the probability of engine lighting. At higher air mass flow rates, the air assist pressure has a greater impact on the probability of engine lighting. Increasing the air assist pressure reduces the probability of lighting. It is surprising that increasing the air assist pressure has a negative impact on ignition as it would be expected to increase the atomization, and thus make combustion more likely. It has, however, been suggested by Wilbraham [13] that the flame is stabilized by the wake formed by the spray, and that if the atomization is promoted excessively, this wake can be disrupted, preventing the flame from stabilizing. It is, consequently, a compromise between promoting atomization to aid ignition and providing sufficiently dense spray efflux to produce sufficient wake to stabilize the flame.

From Fig. 5, it can be seen that over the majority of the SGT-100 ignition window, a 95% start reliability could be achieved with biodiesel at air assist pressures lower that 3 bar. Even for an air assist pressure of 3.21 bar, a very high start reliability could be achieved.

Figure 6 shows the lean ignition limit for biodiesel fuel at air assist pressure of 2.72 bar and the comparison with those of mineral diesel by Martin [14]. The results suggest that at high air flow rates, the biodiesel is easier to ignite than diesel with the ignition being possible at significantly higher AFRs. At lower flow rates, the opposite is true with diesel igniting at higher values of the AFRs. The results also suggest that the lean ignition limit of biodiesel is not sensitive to air temperature, although the scatter at the lowest air flow rate is bigger. Also shown in the figure is the extreme of predicted light up window for the SGT-100 engine for biodiesel. It further suggests that reliable ignition can be achieved for SGT-100 operating with biodiesel.

The air assist pressure has a minimal impact on the lean ignition limit of diesel, although higher air pressure slightly broadens the limit. This also seems to be the case for biodiesel, as shown in Fig. 7, except at low air mass flow rates, where significantly different AFR is required for ignition at different air assist pressures.

The lean extinction limit was also recorded for the biodiesel at air temperatures of 0°C and 30°C, and is illustrated in Fig. 8. This showed that the lean extinction occurred for both cases at the same conditions, and furthermore, that these conditions were similar to those seen for the lean ignition.

The close similarity in the position of the ignition and extinction could indicate that the igniter is placed in a region where combustion is most likely to be sustained. If there had been a

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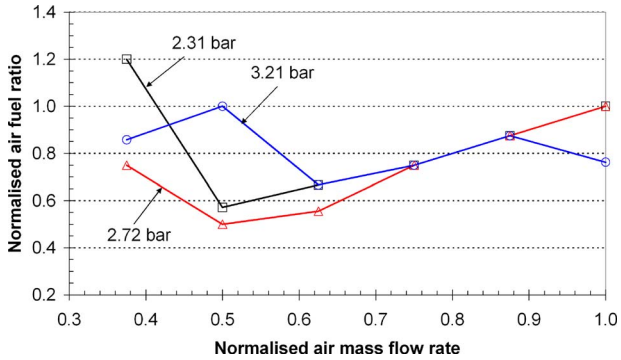

Fig. 7 Effect of air assist pressure on SGT-100 lean ignition limit burning biodiesel at ambient air temperature of 15°C

significant difference between the extinction and ignition curves, it would indicate that the spark was poor or that it was being introduced in an inappropriate place. As the two curves are coincident, within the measurement accuracy available, it indicates that the ignition occurs at the optimal AFR. This in turn would suggest that the design for the igniter position and spark quality for diesel is also optimal for biodiesel.

It is interesting to mention that Wilson et al. [11] showed that their four kinds of biodiesel fuels were harder to ignite than diesel, and that the location of the igniter had a significant impact on the performance. There are several possible reasons for different results from the two different groups. It was found that at the same air mass flow rate, Wilson et al. [11] operated at a much richer fuel condition than Siemens for both fuels. Also, the two groups used different injectors; Siemens used air assist injectors, and Wilson et al. [11] used nonair assist injectors. Different mechanisms for flame stabilization can also contribute to the difference.

5.2 Emissions and Combustion Dynamics. The impact of flame temperature and combustor pressure drop on NO_x emissions are shown in Fig. 9 for two different air inlet pressures of 9 bar and 14 bar at a typical engine combustor full load inlet temperature. Flame temperatures presented in this paper were normalized to the full load flame temperature, and as a result, a normalized value of 1.0 represents the full load condition. At both inlet pressures, diesel had a higher NO_x emission than biodiesel, and, as expected, higher flame temperature resulted in a higher NO_x emission for both fuels. At lower inlet pressure, the combustor pressure drop had an insignificant effect on NO_x formation for both fuels. However, at higher inlet pressure, there was a strong dependence on combustor pressure drop for biodiesel, with the NO_x reduced with the increasing combustor pressure drop. This dependence was not seen for diesel.

It is expected that the NO_x formation would be reduced as the

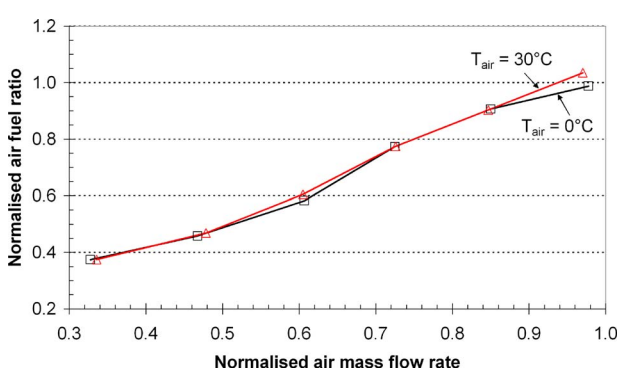

Fig. 8 Effect of ambient air temperature on SGT-100 lean extinction limit burning biodiesel at air assist pressure of 2.72 bar

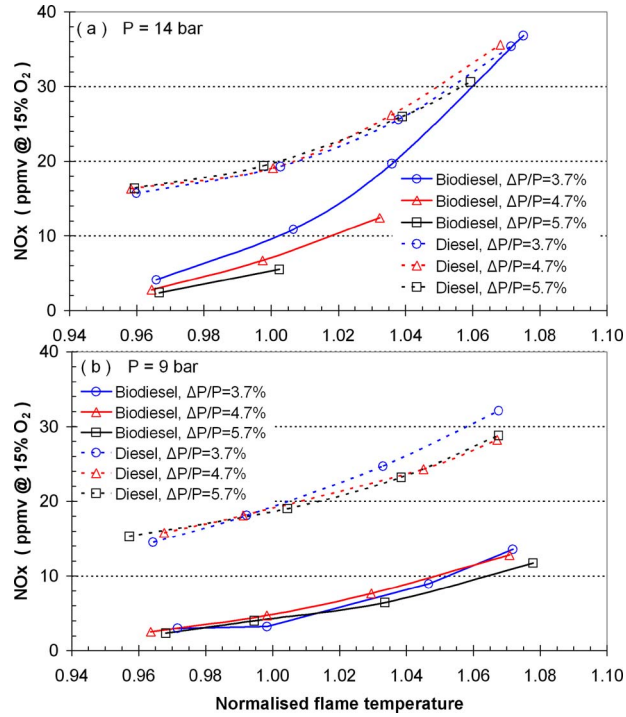

Fig. 9 The effect of the flame temperature and combustor pressure drop on NO_x emission for biodiesel at two different air inlet pressures, and its comparison with diesel

combustor pressure drop is increased for three reasons: (1) the burnt gas residence time in the burner is reduced as the mixture velocity is increased, and as a consequence, the NO_x level is reduced due to the diminished influence of the slow Zeldovich mechanism; (2) particularly for the liquid fuel, as the velocity of the mixture inside the combustor increases, the evaporation and droplet size of the fuel are improved [15], and the mixing with air is enhanced, as a result, NO_x level is reduced; and (3) the mixing of air and fuel is further enhanced by the possible downstream displacement of the flame. However, once perfect mixing is achieved, then the impact of combustor pressure drop on the NO_x emission would be limited to reducing the burnt gas residence time. Syed and Buchanan [16] showed that the NO_x formation is independent of combustor pressure drop, once optimal levels of mixing are achieved in their modified swirler under a gas operation.

Based on the above arguments, for diesel fuel, the optimal level of mixing was achieved in the SGT-100 combustor, regardless of the inlet pressure. However, for biodiesel, at the lower inlet pressure of 9 bar, the optimal level of mixing was achieved, so the combustor pressure drop had a little influence on the NO_x formation, but at the higher inlet pressure of 14 bar, the optimal level of mixing was not achieved, the combustor pressure drop had a great impact on NO_x formation, and mixing was improved by increasing the combustor pressure drop.

As shown in Fig. 9, for biodiesel fuel, comparable levels of NO_x are produced at low flame temperature at the two different inlet pressures, regardless of the combustor pressure drop. As the flame temperature is increased, NO_x formation at higher pressure begins to diverge from those of low pressure, and result in higher NO_x emissions. For diesel fuel, the inlet pressure effect is not significant, although higher pressure produces slightly higher NO_x.

The insignificant dependence of inlet pressure and combustor pressure drop on the NO_x emission for diesel for all the investigated flame temperatures, and for biodiesel at low flame temperature, suggests that well premixed mixtures were achieved, and

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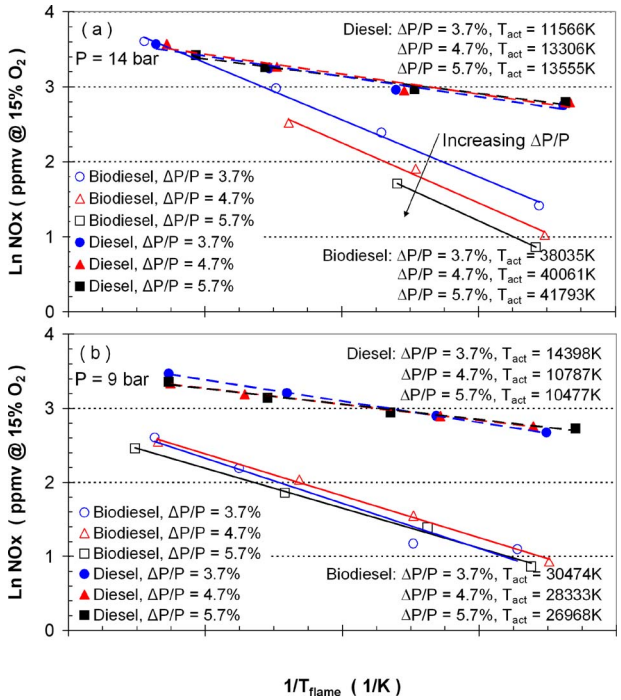

Fig. 10 Activation temperatures for biodiesel and diesel fuel at different air inlet pressures and combustor pressure drops

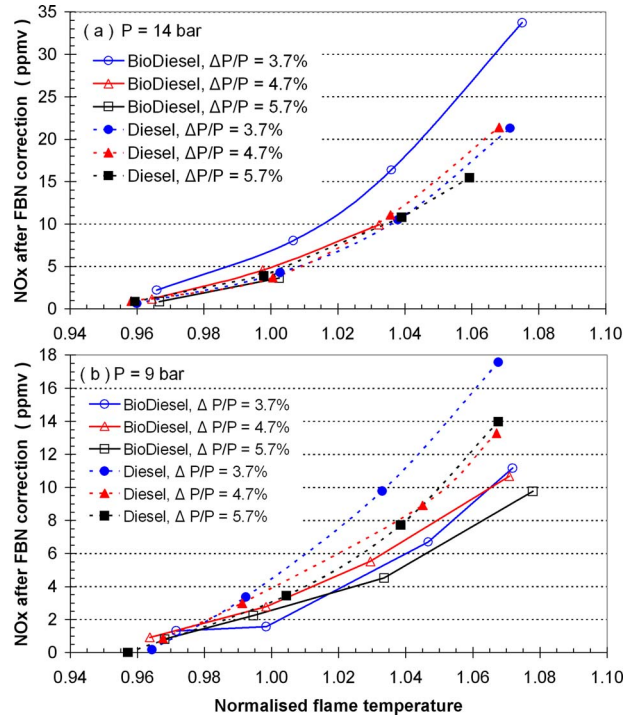

Fig. 11 Comparison of NO_x emissions from biodiesel and diesel after FBN correction

mixing was completed prior to the reaction zone under these conditions. However, for biodiesel, at higher pressure and high flame temperature, as the amount of fuel is increased, there is a clear move away from the minimum level for low pressure. This would suggest that once the amount of fuel supplied exceeds a certain level, mixing is incomplete at the reaction zone.

Figure 9 was replotted as an Arrhenius-type plot in Fig. 10 to get the approximate activation temperature for NO_x formation, based on a simple Arrhenius expression.

Syed and Buchanan [16] demonstrated that the apparent activation temperature is increased as the degree of premixing is increased for gas fuel, and once the optimal mixing level is achieved, the activation temperature is not sensitive to combustor pressure drop. It would be expected that similar behavior would be seen with liquid fuel. This would be complicated by the presence of a fuel bound nitrogen and the NO_x it generates.

As shown in Fig. 10, only for biodiesel at the higher inlet pressure of 14 bar is there a clear trend of activation temperature increasing with the increasing combustor pressure drop. From the link established between activation temperature and mixedness, Fig. 10 further indicates that the degree of premixedness is quite high for diesel at both pressures and for biodiesel at low pressure, but for biodiesel premixedness at higher inlet pressure increases with higher combustor pressure drop. Biodiesel has a higher activation temperature than diesel at the same inlet pressure, and has lower NO_x emissions, which suggests that activation temperature would be an important indicator for NO_x formation.

Campbell et al. [1] and Lapuerta et al. [5] demonstrated that biodiesel fuels have a comparable NO_x emission in industrial gas turbines and internal combustion diesel engines, respectively. This could be because their biodiesel and diesel fuels had comparable nitrogen content. As mentioned before, the biodiesel used in the current study has ten times less nitrogen content than Wilson et al. [11]. Also, the fuel sample analysis showed that the biodiesel used in this study had a significantly less nitrogen content than diesel as listed in Table 1. The experimental study by Eberius et al. [17] and the chemical kinetics study by Warnatz et al. [18] on a propane air flame with 2400 ppm CH_3-NH_2 (methylamine) can be used as a model system for the fuel nitrogen conversion into pollutants. Both studies show that about two-thirds of the fuel nitrogen are oxidized to form NO under fuel lean conditions, and the rest is converted into N_2. Nicol et al. [19] demonstrated that at low equivalence ratios of about 0.5 and 0.6, most of the NO_x is formed by the nitrous oxide mechanism. In the detailed internal measurements on the SGT-100 [20] fueled by mineral diesel, the concentrations of NO and NO_2 were measured. It was found that NO was dominant and was about 80% of the total NO_x. Based on these arguments, an attempt has been made to correct for the fuel bound nitrogen (FBN) effect, assuming that all the nitrogen in the fuel is converted to NO. This correction isolates the influence of FBN, and the effect of mixing and combustion can be identified.

The NO_x emissions corrected for FBN are shown in Fig. 11. A huge reduction in the NO_x level can be seen for diesel, and little reduction is observed for biodiesel. At an inlet pressure of 14 bar, comparable NO_x levels for both fuels can be seen for all flame temperatures and combustor pressure drops, except for biodiesel at the combustor pressure drop of 3.7%, which has the highest NO_x level for all flame temperatures. For an inlet pressure of 9 bar at a lower flame temperature, the NO_x levels are comparable, but diesel shows slightly higher NO_x levels when the flame temperature is increased.

The comparisons of CO emission for both fuels at 15% O_2 are shown in Fig. 12. At an inlet pressure of 14 bar at all flame temperatures investigated, biodiesel has a higher CO emission. This difference is larger at the lowest flame temperature. At the lowest flame temperature, both diesel and biodiesel show strong combustor pressure drop influence on CO emission; CO level is reduced when combustor pressure drop is reduced. When the normalized flame temperature is higher than 0.99, the effect of the combustor pressure drop on the CO level is diminished. At an inlet pressure of 9 bar, the influence of the combustor pressure drop on CO is reduced. Biodiesel has much higher CO emission levels than diesel at the lowest flame temperature, and the difference becomes smaller when the flame temperature is increased until the CO levels for the two fuels are comparable.

For diesel, the effect of the air inlet pressure on CO emission is

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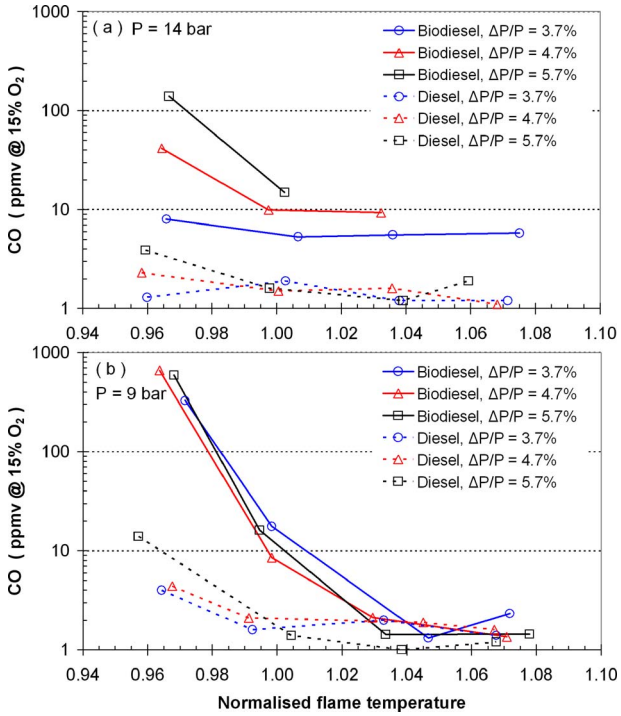

Fig. 12 The effect of flame temperature, combustor pressure drop on CO emission of biodiesel at two inlet pressures, and its comparison with diesel

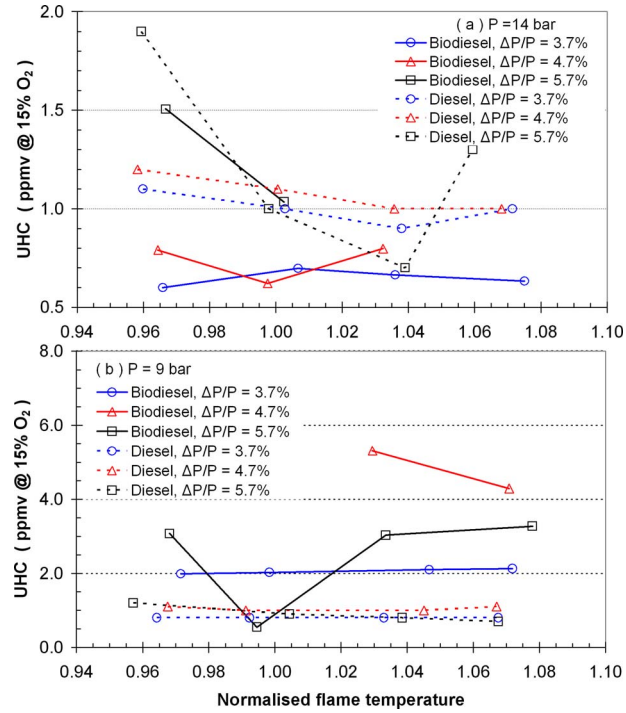

Fig. 13 Comparison of UHC emissions from biodiesel and diesel at two inlet pressures

small when the normalized flame temperature is higher than 0.99. There is some pressure effect at the lower flame temperature, where CO emission is higher when the air inlet pressure is lower. It is interesting to note that for biodiesel at normalized flame temperature of 0.99, there is little pressure effect on CO emission. However, a clear pressure effect is observed at other flame temperatures, with the CO level increased as flame temperature is reduced, and decreased as flame temperature is increased.

The comparisons of unburned hydrocarbon (UHC) emissions are shown in Fig. 13. At a higher air inlet pressure of 14 bar, biodiesel has lower UHC emissions. However, the differences are within 0.5 ppm, and the authors would consider that the two fuels have comparable UHC emissions considering the measurement accuracy. At the lower air inlet pressure of 9 bar, biodiesel has slightly higher UHC emissions, but the differences are within 3 ppm, except at a combustor pressure drop of 4.7%, which has the highest difference of about 4 ppm.

For high pressure rig testing, combustion dynamics were also measured by a pressure transducer mounted at the pilot burner face. The comparisons of combustion dynamic pressure from the two fuels at inlet pressure of 14 bar are shown in Fig. 14 in terms of frequency and root mean square (rms) amplitude. The frequencies and rms pressures were normalized using typical values for the SGT-100 engine at full load conditions. The dynamics frequencies increase as the flame temperature increases for both fuels, but both fuels have comparable dynamics frequencies. Biodiesel has lower rms dynamics pressures at lower flame temperatures, and has higher dynamics when the flame temperatures exceed full load value.

The Sauter mean diameter (SMD) droplet sizes have been estimated to investigate the evaporation process in mixing and combustion. According to Lefebvre [15], the droplet size is a function of dynamic viscosity having $\mathrm{SMD} \propto \mu_L^a$ and a between 0.06 and 0.215. It is also a function of surface tension having $\mathrm{SMD} \propto \sigma^b$ and b about 0.25. The surface tension for biodiesel and diesel were calculated, and the results showed that the surface tension is comparable.

According to Lefebvre [15], SMD can be calculated by Eq. (1) as

$$\mathrm{SMD} = 2.25\sigma^{0.25}\mu_L^{0.25}\dot{m}_L^{0.25}\Delta P_L^{-0.5}\rho_A^{-0.25} \tag{1}$$

where σ is the surface tension (kg/s^2), μ_L is the dynamic viscosity (kg/m s), $\dot{m}_L$ is the liquid mass flow rate (kg/s), ΔP_L is the pres-

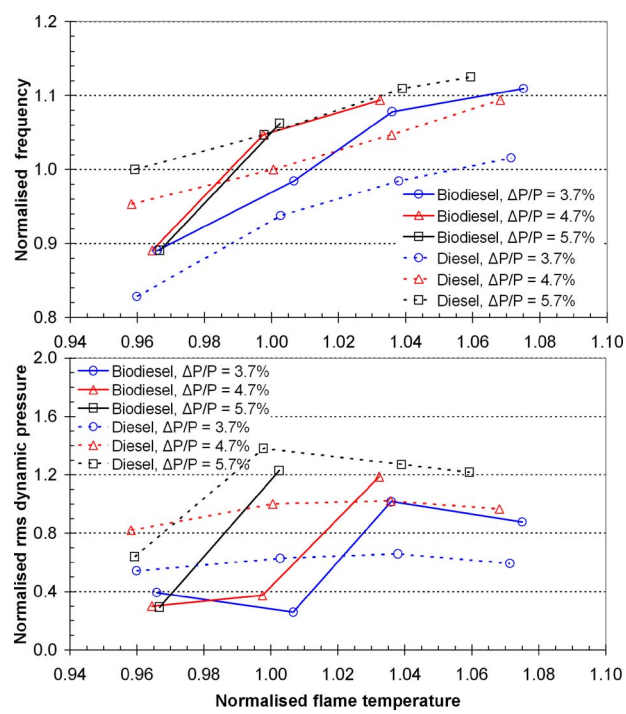

Fig. 14 Comparison of combustion dynamics fueled with biodiesel and diesel at an air inlet pressure of 14 bar

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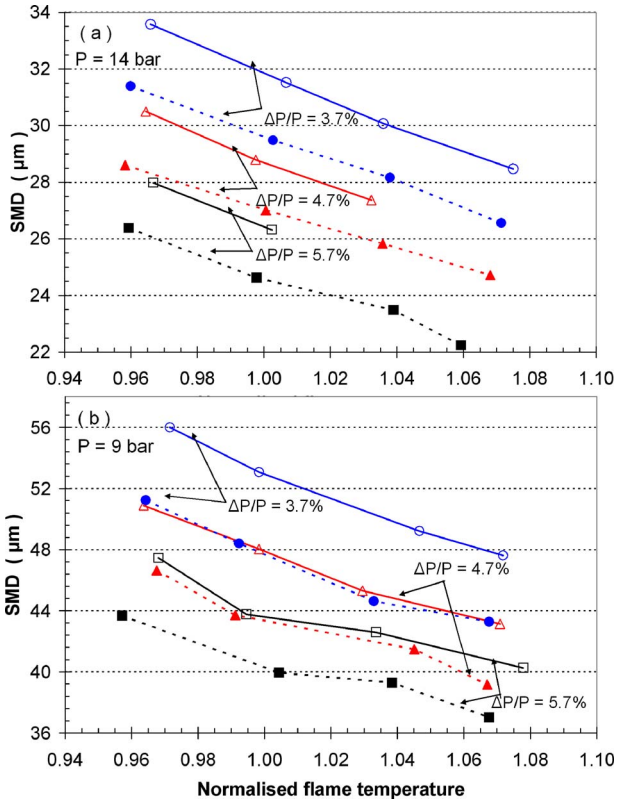

Fig. 15 SMD droplet diameter for biodiesel (solid line) and diesel (dashed line) at different air inlet pressures and combustor pressure drops

sure difference across nozzle (Pa), and ρ_A is the air density (kg/m^3).

The droplet diameters, thus calculated, are shown in Fig. 15. It can be seen that biodiesel has a larger droplet size than diesel at the same condition, and the droplet size is bigger when pressure is reduced for both fuels. According to Eq. (1), the dominant contribution to the different droplet size of both fuels comes from the Dynamic viscosity; as at 313 K, the dynamic viscosities are 4.18×10^{-3} kg/m s and 3.06×10^{-3} kg/m s for biodiesel and diesel, respectively. Biodiesel has a higher dynamic viscosity than diesel, hence, it has a bigger droplet size. It is also significant that evaporation for diesel is better than biodiesel.

Panchasara et al. [9] measured the droplet size of two different biodiesel fuels, and found that the droplet size of biodiesel is bigger than that of diesel. They also demonstrated that the atomization and fuel-air mixing processes have a major impact on NO_x and CO formation. Sequera et al. [10] found that even though the fuel properties are important, the flow effects can dominate the emissions of NO_x and CO, for example, the fraction of the total air used for atomization had a significant effect on both NO_x and CO emission, and emissions decreased with the increase in the atomizing air. They suggest that for a given fuel, the emissions can be minimized by properly tailoring the injector design and the associated combustion processes.

6 Conclusions

Extensive ignition, emissions, and combustion dynamics testing of biodiesel on standard DLE SGT-100 combustion hardware has been conducted to investigate the feasibility of using biodiesel as a fuel to power industrial gas turbines.

In ignition testing, two series of tests were performed to identify the key properties of the proposed fuel. The first set of tests identified the likelihood of the engine lighting, based on the measured probability of the ignition of a single combustor. This demonstrated that a high likelihood of ignition would be possible, depending on the choice of air assist pressure. The second series of tests investigated the lean ignition and extinction limits at various air temperatures. This showed that the ignition limit was different to that measured for mineral diesel, with the biodiesel being easier to ignite at high air mass flow rates, and harder at lower flow rates. It was also demonstrated that air temperature did not have a significant impact on the lean ignition limit. It was further shown that the ignition window for the SGT-100 lays well within the predicted curves for the lean ignition limit, and therefore, good ignition should be possible. The lean extinction limits were found to be very similar to the ignition limits.

Emissions and combustion dynamics were measured in a high pressure test rig. The performance of the burner on the biodiesel was mapped by variation in operating conditions such as the air inlet pressure, temperature, combustor pressure drop, and flame temperature. The results were compared with those of diesel at the same conditions. For all the operating conditions tested, biodiesel had a lower NO_x emission than diesel. For biodiesel at an air inlet pressure of 14 bar, there was a strong link between NO_x and the combustor pressure drop, where the NO_x level was reduced as the combustor pressure drop was increased. It was not the case for the biodiesel at 9 bar, and for the mineral diesel at all the pressures investigated. However, after the FBN correction, comparable NO_x emissions were obtained, although diesel showed a slightly higher NO_x emission than biodiesel at 9 bar inlet pressure.

The analysis of the high pressure rig testing results showed that for biodiesel at low flame temperature and low air inlet pressure, a high degree of premixing was achieved, so NO_x level was independent of the inlet pressure and combustor pressure drop. At high pressure and high flame temperature, the optimal level of mixing could not be achieved due to the amount of fuel supplied. Where mixing is not optimal, the thermal NO_x formation is higher and dependent on the inlet pressure and combustor pressure drop. However, for mineral diesel, good premixing was achieved at all the pressures investigated, and little dependence on the inlet pressure and combustor pressure drop was observed.

At lower flame temperature, biodiesel had significantly higher level of CO than diesel, particularly at a lower inlet pressure of 9 bar. For both fuels, the combustor pressure drop had apparent influence on CO at lower flame temperature with CO level decreasing with the combustor pressure drop decrease, and this influence diminished as the flame temperature was increased.

Both fuels had a comparable level of UHC emission at higher inlet pressure, and biodiesel had slightly higher UHC emissions at lower inlet pressure.

For combustion dynamics, both fuels had comparable frequencies, but biodiesel had lower rms dynamic pressures at lower flame temperatures, and higher when flame temperatures exceed the full load value.

Fuel droplet sizes were calculated, and biodiesel had a bigger droplet size, particularly at lower pressure, lower flame temperature, and lower combustor pressure drop. The dominant contribution of big droplet size for biodiesel came from its much higher viscosity.

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